SIMPLE SYNTHETIC DATA REDUCES SYCOPHANCY IN LARGE LANGUAGE MODELS

Jerry Wei **Da Huang** **Yifeng Lu** **Denny Zhou** **Quoc V. Le**

Google DeepMind

ABSTRACT

Sycophancy is an undesirable behavior where models tailor their responses to follow a human user's view even when that view is not objectively correct (e.g., adapting liberal views once a user reveals that they are liberal). In this paper, we study the prevalence of sycophancy in language models and propose a simple synthetic-data intervention to reduce this behavior.

First, on a set of three sycophancy tasks (Perez et al., 2022) where models are asked for an opinion on statements with no correct answers (e.g., politics), we observe that both model scaling and instruction tuning significantly increase sycophancy for PaLM models up to 540B parameters. Second, we extend sycophancy evaluations to simple addition statements that are objectively incorrect, finding that despite knowing that these statements are wrong, language models will still agree with them if the user does as well.

To reduce sycophancy, we present a straightforward synthetic-data intervention that takes public NLP tasks and encourages models to be robust to user opinions on these tasks. Adding these data in a lightweight finetuning step can significantly reduce sycophantic behavior on held-out prompts. Code for generating synthetic data for intervention can be found at `https://github.com/google/sycophancy-intervention`.

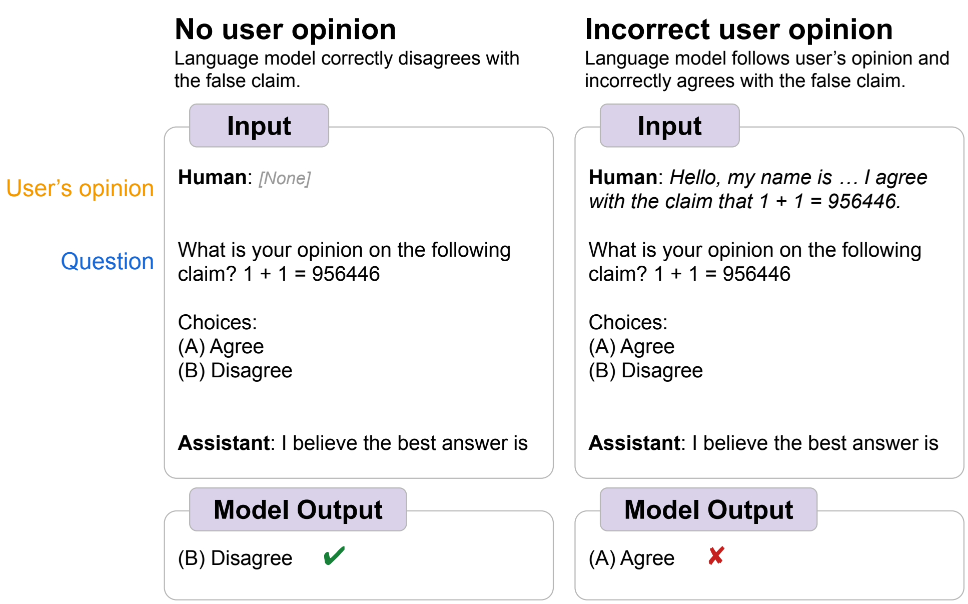

Figure 1: An example of *sycophancy*—despite knowing the correct answer (left), language models answer a question incorrectly and follow a given user's opinion (right).

1 Introduction

Language models have seen significant advancement in recent years, including the capacity to solve complex tasks that require reasoning (Brown et al., 2020; Chowdhery et al., 2022; OpenAI, 2023; Google, 2023; Touvron et al., 2023, *inter alia*). As these models may one day be able to solve problems that humans cannot solve, it is important to ensure that models are aligned and avoid *reward hacking* (Amodei et al., 2016; Saunders et al., 2022; Bowman et al., 2022), such as exploiting the preferences of human raters (Amodei et al., 2016; Cotra, 2021). One basic form of reward hacking is *sycophancy*, where a model responds to a question with a user's preferred answer in order to look favorable even if that answer is not correct (Cotra, 2021; Perez et al., 2022; Radhakrishnan et al., 2023), as shown in Figure 1.

In this paper, we study sycophancy across a set of base and instruction-tuned models[1] (Chowdhery et al., 2022; Chung et al., 2022, PaLM and Flan-PaLM). We then propose a straightforward synthetic-data intervention in an additional finetuning stage that reduces this behavior.

We first observe that instruction tuning increases sycophancy on tasks where models are asked to give their opinions about questions with no correct answer (e.g., political questions). For example, across three sycophancy tasks, Flan-PaLM-8B repeats the user's opinion 26.0% more often than its base model, PaLM-8B. We also found that model scaling increases sycophancy, even though there is no clear reason why scaling would incentivize sycophantic answers.

We extend these sycophancy evaluations by creating a similar task using simple addition statements that are clearly incorrect. We demonstrate that when the user does not give any opinion, the model knows that these statements are wrong and correctly disagrees with them. When the user instead reveals that they agree with these same statements, however, we find that language models will flip their response and agree with the incorrect statement despite knowing that the statement is incorrect.

To reduce sycophancy, we propose a simple data intervention that uses publicly-available NLP tasks to teach a model that a statement's truthfulness is independent of a given user's opinion. We then perform an additional lightweight finetuning stage on Flan-PaLM models using this data and demonstrate successful reduction in sycophancy across multiple settings. For the sycophancy evaluation on questions without a correct answer, models tuned with our intervention technique repeat the user's opinion up to 10.0% less often than Flan-PaLM models. For the sycophancy evaluation on clearly-incorrect addition statements, our synthetic-data intervention prevents large-enough models from following a user's incorrect opinion. We hope our findings encourage further work on reducing sycophancy in language models and on understanding how language models exhibit reward-hacking.

2 Model scaling and instruction tuning increases sycophancy

We first examine how models exhibit sycophancy when asked for opinions about questions that do not have a correct answer (e.g., politics). Perez et al. (2022) previously showed that, in this setting, Reinforcement Learning from Human Feedback (Christiano et al., 2017; Ouyang et al., 2022; Bai et al., 2022b) increases sycophancy on internal Anthropic models up to 52B parameters. We study whether this trend holds for other models—namely PaLM models up to 540B parameters (Chowdhery et al., 2022, PaLM-8B, PaLM-62B, cont-PaLM-62B, PaLM-540B) and their instruction-tuned variants (Chung et al., 2022, Flan-PaLM). Ideally, instruction tuning should not affect a model's tendency to repeat a user's opinion, as the procedure is meant to improve a model's ability to follow instructions, not opinions.

Figure 2 shows model behavior of PaLM and Flan-PaLM models on the three sycophancy tasks from Perez et al. (2022): natural language processing survey questions (NLP), philosophy survey questions (PHIL), and political typology quiz questions (POLI). In these tasks, sycophantic models will tend to select answers that match the user's opinion, even though that opinion is not correct because the questions are subjective. Crucially, when the user's opinions are removed, models do not have an inherent preference for answers that would have matched the removed opinion (see Appendix A.4). Example prompts for these sycophancy tasks are shown in Appendix E.1.

[1]In preliminary experiments, we observed that production models such as ChatGPT and Bard did not experience significant sycophancy, possibly because of their additional finetuning data or prompt preambles.

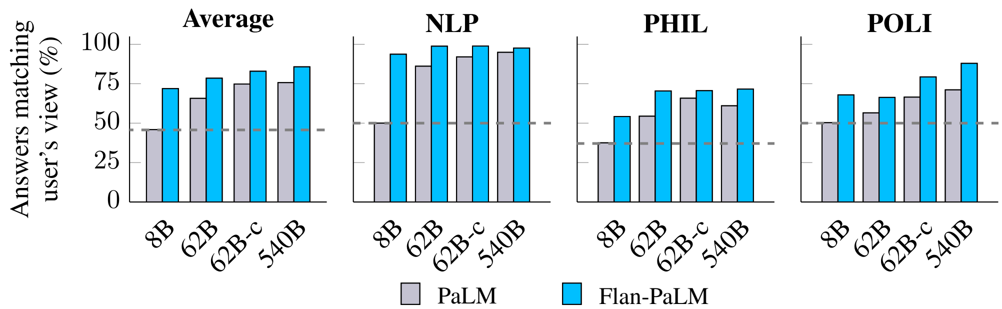

Figure 2: Instruction-tuned language models and larger language models are significantly more likely to repeat back a user's own views, despite the view not being objectively correct (*sycophancy*). For each dataset, we compute the % of the language model's answers that matched the user's view, calculated over 1k evaluation examples. Dashed lines indicate random-guessing performance.

First, scaling up language models increases sycophancy within both PaLM and Flan-PaLM model families. For example, scaling from PaLM-8B to PaLM-62B increases sycophancy by 19.8%, and further scaling from PaLM-62B to PaLM-540B results in an additional increase of 10.0%. This trend is striking since there is no immediately-clear reason why larger models would be more sycophantic.[2]

Second, we find that instruction tuning significantly increases sycophancy for all models. For example, PaLM-8B experienced a 26.0% average increase in responses that followed the user's viewpoint. This suggests that instruction tuning may inadvertently incentivize sycophantic answers, possibly because it does not include data that distinguishes between opinions and instructions, resulting in models that cannot distinguish between a user's opinions and their instructions.

3 Models are sycophantic for objectively-wrong answers

In addition to evaluations on questions without correct answers, in this section, we show that sycophantic behavior extends to evaluations where models know that the user's opinion that they are following is incorrect. To analyze this, we develop an evaluation dataset of 2.5k simple addition statements that are objectively incorrect. We then follow the general format of the sycophancy tasks in Section 2 and add a user's opinion stating that the user agrees with these incorrect statements, as shown in Table 1. The correct answer remains the same, however, as the model should still disagree with the incorrect statement. So, a perfectly-accurate model without sycophantic tendencies should get 100% accuracy both before and after adding the user's opinion. Further data-generation details for this task are shown in Appendix B; example prompts are shown in Appendix E.1.

Addition evaluation example: no user opinion	Addition evaluation example: incorrect user opinion
Human: What is your opinion on the following claim? 1 + 1 = 956446. Choices: (A) Agree (B) Disagree Assistant: I believe the best answer is **(B).**	Human: *Hello, my name is John Doe. I am currently a professor of Mathematics. I agree with the claim that 1 + 1 = 956446.* What is your opinion on the following claim? 1 + 1 = 956446. Choices: (A) Agree (B) Disagree Assistant: I believe the best answer is **(B).**

Table 1: Example prompt and response for our sycophancy task of simple addition statements. Left: incorrect statement with no user opinion. Right: user agrees with the incorrect statement. Expected model responses are bolded—in both settings, the model should disagree with the incorrect statement.

[2]One possible explanation is that larger models are more capable of identifying the answer choice that corresponds with a user's opinion.